Today's Assignment

- Special Index → Prefix sum.
- Even odd Sum.
- Majority Element.

Ques¹ Given N array elements. Find majority element

↳ an element with freq > N/2
i.e. more than half.

N

e.g. arr[6] = 1 2 1 6 1 1

freq(1) = 4 4 > 3 $\boxed{ans = 1}$
N/2 = 3

e.g. arr[9] = 3 4 4 8 4 9 4 3 4

freq(4) = 5 5 > 4 ⟶ ans = 4
N/2 = 9/2 = 4

e.g. arr[11] = 3 3 4 6 1 3 2 5 3 3 3

freq(3) = 6 , N/2 = 11/2 = 5

ans = 3

e.g. arr[10] = 4 6 5 3 4 5 6 4 4 4

freq(4) = 5 , N/2 = 5

5 > 5

You don't have majority element here. $\boxed{ans = -1}$

Approaches

1) For every element, get the frequency and the compare it with $N/2$.

$T.C = O(N^2)$; $S.C = O(1)$ → using nested loop

$T.C = O(N)$; $S.C = O(N)$ → using HashMap → Do not worry.

2) Sorting the array.

Sort the array and get frequency of every element and get the element with freq > $N/2$

e.g. 3 3 4 6 1 3 2 5 3 3 3

sort :- 1 2 3 3 3 3 3 3 4 5 6

freq :- 1 1 6

break and return 3.

$T.C = (N\log N + N)$

$S.C = O(1)$

Hint :- Given N array elements

1) freq (ele) > $N/2$

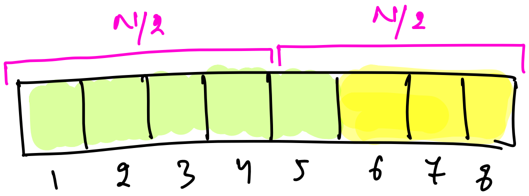
→ Assume 2 majority elements, ele 1 and ele 2

freq (ele1) > $N/2$

freq (ele 2) > $N/2$

freq (ele1) + freq (ele 2) > N → This is impossible.

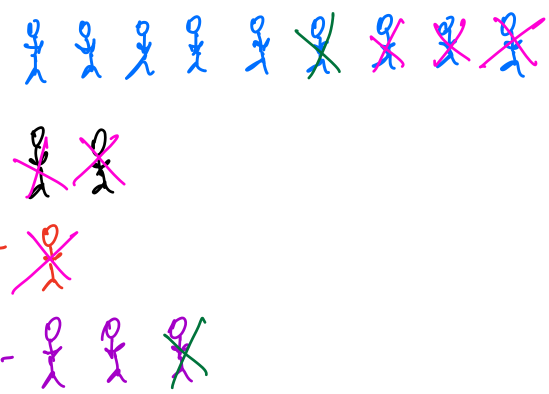
e.g.



Idea:- (i) At max, we can have only 1 majority element
(ii) Its freq. is greater than all other elements frequencies combined.

BJP / Congress / AAP / Kalyani → 15 MLA's → U.P.

BJP :-
AAP :-
Congress :-
Kalyani :-



total = 15

$9 > 15/2$

BJP is majority here

total = 13

$8 > 13/2$

Majority is still BJP

total = 11

$7 > 11/2$ Still BJP for majority

↓

$6 > 9/2$

↓

$5 > 7/2$

Observation :-
If we remove 2 distinct things, the majority will still remain the same.

e.g.

0	1	2	3	4	5	6	7	8
3	4	4	8	4	9	4	3	4

N	N/2	majority element.
9	freq(4) > 9/2	4
7	freq(4) > 7/2	4
5	freq(4) > 5/2	4
3	freq(4) > 3/2	4
1	freq(4) > 1/2	4

1) How to randomly pick 2 different elements.
2) How can you delete them
3) How can you get the frequencies.

Algorithm → Moore's Voting Algorithm. → Google Interviews.
→ Arcesium

→ 2 variables
- ele
- freq.

e.g.

	0	1	2	3	4	5	6	7	8
	3	4	4	8	4	9	4	3	4
ele =	3		4		4		4		4
freq =	1	0	1	0	1	0	1	0	1

4 → final ans

* When different elements are compared to curr ele, reduce the curr ele freq.
* When same element as our curr ele, increase the freq. of curr ele.

e.g.

	0	1	2	3	4	5	6	7	8	9	10
arr[] =	3	3	4	6	1	3	2	5	3	3	3
ele =	3	3	3		1		2		3	3	3
freq =	1	2	1	0	1	0	1	0	1	2	3

$\boxed{\text{Ans} = 3}$

e.g.

arr[] :-	4	6	5	3	4	5	6	4	4	4
ele :-	4		5		4		6		4	4
freq :-	1	0	1	0	1	0	1	0	1	2

freq(4) = 5
N/2 = 5 ✗

4 is not majority element.

* Imp
Always iterate and count freq and compare with N/2.

e.g.

	4	4	4	6	6	6	7
ele =	4	4	4	4	4	4	7
freq =	1	2	3	2	1	0	1

You will iterate and get freq. of 7.
check if it is > N/2

e.g.

arr[7] =	3	4	3	3	6	3	7
ele =	3		3	3	3	3	3
freq =	1	0	1	2	1	2	1

freq(3) = 4

$N/2 = 7/2 = 3$

$4 > 3$

ans = 3

// Pseudo Code.

```
ele = arr[0]; freq = 1
for(i=1; i<N; i++){
    if (freq==0) { ele = arr[i]; freq=1 };
    else if (ele != arr[i]){ freq--; } → //deleting 2 distinct elements here
    else { freq ++ };
}
int c=0;
for (i=0; i<N; i++){
    if(ele == arr[i]) { c++; }
}
if (c > (N/2)) return ele;
else return -1;
```

T.C = O(N)

S.C = O(1)

TODO:- Given N array elements. Check if there exists freq(ele) > N/3

Majority ele = an element with freq(ele) > N/3

→ How, many majority elements can we have?

1) $N/3$, $N/3$

$2N/3 < N$

HINT:- ele 1, freq1 ↓ Majority element 1   ele 2, freq 2 ↓ Majority ele 2.

2) If we remove 3 distinct elements, the majority won't change